Multiclass Classification of Cardiac Arrhythmias Using Two-Dimensional Convolutional Neural Networks and ECG Spectro-Temporal Representations

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***Abstract*—This article presents a comprehensive approach for the automatic classification of cardiac arrhythmias using two-dimensional convolutional neural networks (CNNs) on time-frequency representations of electrocardiogram (ECG) signals. The study specifically addresses the problem of extreme imbalance in clinical datasets and the loss of spectral information in conventional one-dimensional models. The proposed methodology includes four main phases: (1) ECG signal preprocessing with R-peak-centered segmentation and baseline removal via polynomial fitting; (2) transformation to the time-frequency domain using the Continuous Wavelet Transform (CWT) with the complex Morlet wavelet; (3) implementation of controlled balancing and data augmentation strategies to mitigate extreme class imbalance; and (4) design and optimization of a custom CNN architecture specifically for ECG spectrograms. Evaluation was performed on the MIT-BIH Arrhythmia Database, considering five beat classes: Normal (N), Supraventricular Ectopic (S), Ventricular Ectopic (V), Fusion (F), and Unknown (Q). The main model, trained with balanced data, achieved an overall accuracy of 99.70%, a macro F1 of 99.70%, and an average AUC-ROC of 99.99%, significantly outperforming results reported in the state of the art. A controlled experiment with the same architecture but without balancing techniques showed consistently lower performance (accuracy: 98.29%, macro F1: 93.75%), demonstrating the critical impact of balancing strategies on minority-class classification. The results demonstrate the feasibility of the proposed approach for clinical applications and highlight the importance of spectro-temporal representations and systematic balancing strategies in AI-assisted diagnostic systems.**

***Index Terms*—Arrhythmia classification, ECG, convolutional neural networks, continuous wavelet transform, data balancing, deep learning, time-frequency representations.**

I. Introduction

Cardiovascular diseases represent one of the main global health problems, significantly affecting the quality of life of those who suffer from them. According to data from the Pan American Health Organization (PAHO), in the Americas region there has been a decrease in the age-standardized cardiovascular mortality rate, from 203.3 deaths per 100,000 inhabitants in 2000 to 137.2 in 2019 [1]. However, accurate and timely diagnosis of cardiac arrhythmias remains a significant clinical challenge.

The electrocardiogram (ECG) is the fundamental tool for cardiac diagnosis, although it presents important challenges due to its non-stationary nature and inter-patient variability [2]. Manual evaluation of ECG recordings remains a complex process, especially when large volumes of data from multiple patients must be analyzed. In this context, automatic ECG signal classification emerges as a promising alternative to support healthcare professionals in the accurate diagnosis of various cardiac pathologies.

Traditional approaches for ECG signal classification have included Bayesian methods [37], heuristic methods [38], expert systems [39], and Markov models [40]. However, these methods present important limitations related to their sensitivity to noise and difficulties in generalizing to novel or ambiguous patterns [41]. In contrast, deep learning-based approaches, particularly convolutional neural networks (CNNs), have demonstrated superiority in automatically extracting representative features from ECG signals [5].

However, existing models face several specific challenges:

1) **Extreme imbalance**: In real arrhythmia datasets, the clinically most critical classes (such as supraventricular and ventricular beats) are significantly less frequent than normal beats.
2) **Loss of spectral information**: Conventional one-dimensional CNN architectures operate exclusively in the time domain, ignoring valuable frequency information present in ECG signals.
3) **Overfitting in minority classes**: Non-optimized architectures tend to prioritize majority classes during training, resulting in poor performance for clinically relevant but infrequent classes.
4) **Incomplete evaluation**: The use of global metrics such as accuracy can hide poor performance in minority classes.

This work proposes a comprehensive system that addresses these limitations through the following main contributions:

1) Implementation of time-frequency representations using the Continuous Wavelet Transform (CWT) to capture spectral and temporal information simultaneously.

Table I: Comparative summary of the state of the art in arrhythmia classification

Study	Approach	Acc. (%)	F1 (%)	AUC (%)	Year
Osowski et al. (2004) [3]	SVM + Hermite coefficients	89.3	78.9	94.2	2004
de Chazal et al. (2004) [4]	Linear discriminant analysis	85.9	–	–	2004
Chourasia et al. (2020) [5]	Basic CNN	96.2	95.1	98.3	2020
Jun et al. (2018) [6]	2D-CNN + STFT	94.1	85.7	97.5	2018
Pramudito et al. (2022) [7]	1D-CNN	94.1	85.7	97.5	2022
Our study	**2D-CNN + CWT + balancing**	**99.7**	**99.7**	**99.99**	**2024**

2) Design of controlled balancing and data augmentation strategies specifically adapted for medical images of ECG spectrograms.
3) Development of a custom CNN architecture optimized for the specific characteristics of ECG spectrograms.
4) Rigorous evaluation through stratified cross-validation and class-specific metrics, including a control experiment to quantify the impact of balancing.

II. Related Work

Table I presents a comparative summary of the main approaches reported in the literature for cardiac arrhythmia classification:

A. Approaches Based on Handcrafted Features

Traditional methods for arrhythmia classification have been based on manual extraction of morphological and temporal features from ECG signals. Osowski et al. [3] proposed an expert system based on support vector machines (SVM) using Hermite transform coefficients. Although this approach demonstrated robustness, it requires specialized knowledge for descriptor extraction, which introduces subjectivity and limits its adaptability to signal variations. Similarly, de Chazal et al. [4] developed a linear discriminant analysis classifier independent for each feature, although they were limited to a binary classification approach that lacks the capacity necessary to address real clinical scenarios in which multiple types of arrhythmias coexist.

B. Deep Learning-Based Approaches

Deep learning architectures, particularly convolutional neural networks (CNNs), have revolutionized the field of biomedical signal classification. Chourasia et al. [5] implemented a basic CNN architecture for multiclass heart beat classification, achieving an accuracy of 96.2%. However, their approach used a generic architecture without specific optimizations for ECG characteristics. Jun et al. [6] explored the use of two-dimensional CNNs on time-frequency representations generated by the Short-Time Fourier Transform (STFT), but their approach was limited to a single ECG lead and showed high variability across different data partitions.

C. Identified Limitations in the Literature

Analysis of the state of the art reveals several methodological limitations:

1) **Unaddressed imbalance**: Most previous studies did not systematically address the extreme imbalance inherent in arrhythmia datasets.
2) **Suboptimal representation**: One-dimensional approaches lose valuable spectral information, while existing time-frequency representations (such as STFT) have fixed resolution.
3) **Generic architectures**: The use of standard CNN architectures without specific adaptations for the ECG signal domain.
4) **Incomplete evaluation**: Excessive dependence on global metrics that hide poor performance in minority classes.

III. Methodology

The development of this research followed a quantitative experimental approach, based on the implementation and comparative evaluation of deep learning models for multiclass ECG signal classification. The methodological process was structured into four sequential and integrated phases that included preprocessing and spectral representation of signals, exploratory experimentation with transfer learning, balancing and data augmentation strategies, and the design, optimization, and evaluation of a custom CNN architecture.

A. Database

1) MIT-BIH Arrhythmia Database: The MIT-BIH Arrhythmia Database was used [8], considered the international standard reference for validating arrhythmia detection algorithms. Developed jointly by Beth Israel Hospital and the Massachusetts Institute of Technology, this database contains 48 ECG recordings of 30 minutes each, sampled at 360 Hz. Each recording includes two ECG leads (MLII and V5), selected for their complementarity in capturing different aspects of cardiac electrical activity.

2) Class Distribution: Table II shows the initial distribution of beat types in the dataset:

Table II: Initial distribution of beat types in MIT-BIH

Beat Type	Number	Percentage
Normal (N)	150,680	68.5%
Supraventricular Ectopic (S)	5,428	2.5%
Ventricular Ectopic (V)	28,980	13.2%
Fusion (F)	3,568	1.6%
Unknown (Q)	36,494	16.6%
Total	**225,150**	**100%**

3) Description of Classes:

- **Normal Beats (N)**: Represent normal cardiac rhythm (sinus rhythm) with characteristic morphology: well-defined P wave, narrow QRS complex (¡120 ms), and concordant T wave.
- **Supraventricular Ectopic Beats (S)**: These are premature beats that originate above the ventricles (atria or AV junction). They typically present narrow QRS complexes, but with abnormal PR interval and incomplete compensatory pause.
- **Ventricular Ectopic Beats (V)**: These are premature beats that originate in the ventricles. They present wide and deformed QRS complexes (¿120 ms), with discordant ST segment and T wave.
- **Fusion Beats (F)**: Occur when a normal sinus impulse and an ectopic ventricular impulse depolarize the ventricles simultaneously. They present hybrid morphology between a normal and a ventricular beat.
- **Unknown Beats (Q)**: Include beats that cannot be classified into the previous categories, beats with unspecified aberrant morphology, and beats whose annotation in the database is ambiguous.

B. ECG Signal Preprocessing

1) Beat Segmentation: Each beat was segmented using a 0.6-second window (216 samples) centered on the R peak, according to the methodology established by Jun et al. [6]. This temporal window was selected after an exhaustive analysis of the literature, allowing adequate capture of the main components of the cardiac beat: the P wave, the QRS complex, and the T wave. The window of 0.2 seconds before the R peak captures atrial activity (P wave) and the beginning of ventricular depolarization, while the 0.4 seconds after capture ventricular repolarization (T wave).

The segmentation equation is:

$$\text{window} = [\text{R_peak} - 0.2 \times f_s, \text{R_peak} + 0.4 \times f_s] \quad (1)$$

where $f_s = 360$ Hz is the sampling frequency.

2) Baseline Removal: ECG signals acquired in clinical environments often present baseline wander, a low-frequency artifact caused by factors such as patient respiration, body movement, or imperfections in electrode coupling. This artifact can mask important morphological features of the ECG and significantly affect the performance of classification algorithms.

A sixth-order polynomial fit was applied to remove this deviation [9], a technique that has demonstrated superiority over traditional methods such as high-pass filters. The sixth degree of the polynomial was selected through experimental validation, finding that it provides an optimal balance between the ability to model low-frequency trends and the preservation of signal components of interest.

The correction is implemented as:

$$s_{\text{corrected}}(t) = s_{\text{original}}(t) - \sum_{k=0}^{6} a_k t^k \quad (2)$$

where a_k are the polynomial coefficients obtained by least squares:

$$\min_{a_k} \sum_{t=1}^{N} \left(s_{\text{original}}(t) - \sum_{k=0}^{6} a_k t^k \right)^2 \quad (3)$$

3) Continuous Wavelet Transform (CWT): The generation of spectrograms using the Continuous Wavelet Transform represented a significant innovation in this work, allowing one-dimensional ECG signals to be transformed into two-dimensional representations rich in time-frequency information. The complex Morlet wavelet was used [34] due to its mathematical properties, which make it particularly suitable for analyzing biological signals: it provides an optimal balance between temporal and frequency resolution, and its complex nature allows simultaneous capture of phase and amplitude information.

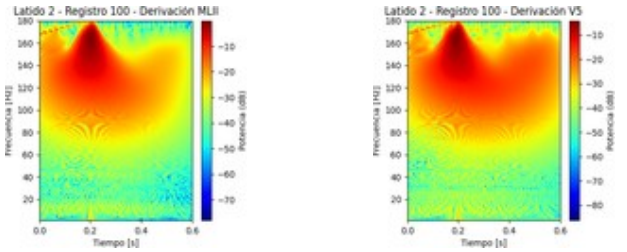
The selection of CWT parameters was performed through an iterative optimization process, considering computational aspects and the preservation of clinically relevant characteristics. The scales were selected to cover the frequency range of interest in the ECG (0.5-40 Hz), while the spatial resolution of 128×128 pixels represents a compromise between representational detail and computational efficiency.

The CWT is defined as:

$$\text{CWT}(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} s(t) \psi^* \left(\frac{t - b}{a} \right) dt \quad (4)$$

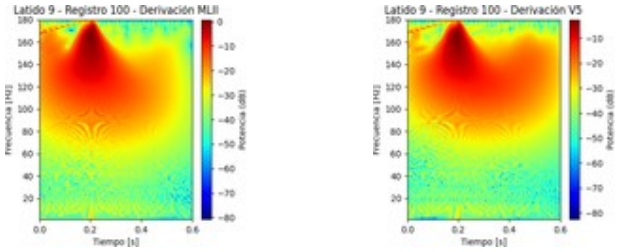
where $\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-t^2/2}$ is the Morlet wavelet, with $\omega_0 = 6$ rad/s for an optimal balance between temporal and frequency resolution.

Figure 1 shows representative examples of spectrograms generated for each class.



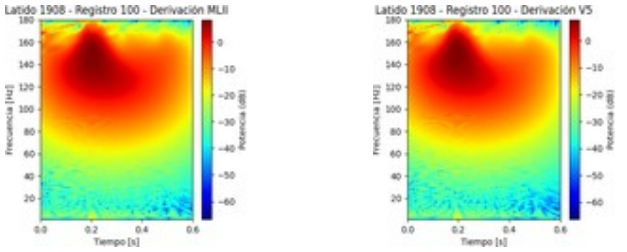
MLII **V5**

(a) Normal Beat: Exhibits a regular rhythmic pattern with well-defined spectral components and characteristic harmonic distribution.



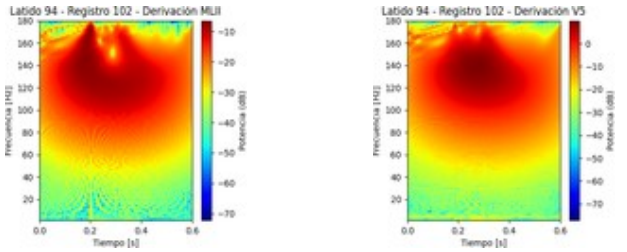
MLII **V5**

(b) Supraventricular Beat: Shows alterations in fundamental frequencies with relative preservation of temporal structure.



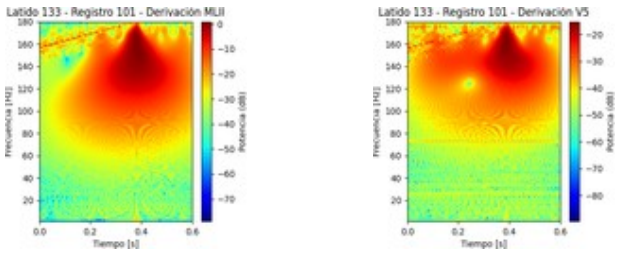
MLII **V5**

(c) Ventricular Beat: Presents QRS complexes spectrally expanded and alterations in energy distribution.



MLII **V5**

(d) Fusion Beat: Combines spectral characteristics of normal and ventricular beats, showing hybrid patterns.



MLII **V5**

(e) Unknown Beat: Exhibits atypical patterns with irregular and non-harmonic spectral distributions.

Figure 1: Representative examples of spectrograms by class and by lead

C. Exploratory Experimentation with Transfer Learning

1) Objective and Experimental Design: This phase aimed to determine the optimal ECG lead configuration through a rigorous comparative evaluation of three different configurations. The selection of leads directly affects the amount and quality of information available to the model, which affects its generalization capacity and diagnostic performance.

The experimental design considered three progressively more complex configurations:

1) **Individual MLII model**: Used only the MLII lead.
2) **Individual V5 model**: Used only the V5 lead.
3) **Fusion model**: Combined both leads by concatenating features.

2) MobileNetV2 Architecture with Transfer Learning: The selection of MobileNetV2 as the base architecture for this exploratory phase was based on its computational efficiency and its inverted residual block architecture, which facilitates gradient flow during training [11]. The use of transfer learning from ImageNet allows leveraging generic visual representations learned from a massive dataset, which can be adapted to the specific domain of ECG spectrograms.

The fine-tuning process was implemented through a strategy of fully freezing the base network, keeping all pretrained MobileNetV2 parameters unchanged during training. This feature extraction approach is particularly suitable for limited datasets, as it preserves the hierarchical visual representations learned on ImageNet while avoiding overfitting. Only the final classifier layer was replaced and trained, adapting the architecture from the original 1,000 classes to the 5 target classes of the heart beat classification problem.

For individual models:

$$y = \text{Softmax}\left(\text{Linear}\left(\text{Dropout}\left(\text{Features}_{\text{MobileNetV2}}(x_{\text{lead}})\right)\right)\right) \tag{5}$$

For the fusion model:

$$\begin{aligned} y = &\text{Softmax}\Big(\text{Lin}\big(\text{Drop}\big(\\ &\text{ReLU}\big(\text{Lin}_{512}[F_{\text{MLII}} \oplus F_{\text{V5}}]\big)\big)\big)\Big) \end{aligned} \tag{6}$$

$\text{Lin} = \text{Linear}$, $\text{Drop} = \text{Dropout}$, $F_{\text{MLII/V5}} = \text{Features}_{\text{MLII/V5}}$

3) Results of the Exploratory Phase: Table III presents the results of the exploratory phase with MobileNetV2:

Table III: Results of the exploratory phase with MobileNetV2

Configuration	Overall Acc.	Macro F1	AUC-ROC	Average AP
Individual MLII	91.99%	73.52%	95.34%	79.01%
Individual V5	89.59%	72.18%	94.86%	76.78%
Fusion (MLII+V5)	**97.87%**	**92.09%**	**99.26%**	**94.99%**

The results consistently showed that the fusion configuration with both leads significantly maximized performance in all evaluated metrics. The improvement, particularly notable in macro F1 (an increase of approximately 18.5 percentage points compared with the individual MLII model), indicates that the fusion model handles the inherent class imbalance substantially better, distributing its predictive capacity more equitably

among all categories. This finding justified the adoption of the fusion strategy in all subsequent phases of the project.

D. Balancing and Data Augmentation

1) Initial Imbalance Analysis: The initial analysis of the dataset revealed extreme imbalance: the Normal class represented 68.5% of the total, while clinically critical classes, such as supraventricular beats, represented only 2.5%. This imbalance represents a significant challenge for machine learning algorithms, which tend to produce models that prioritize the majority class to the detriment of minority classes, which is clinically unacceptable, given that the most dangerous arrhythmias are often less frequent.

2) Controlled Augmentation Techniques: Data augmentation techniques validated specifically for medical images were implemented [12], designed to increase dataset diversity while preserving clinically relevant characteristics of the spectrograms. Each technique was applied in a controlled manner to avoid introducing artifacts that could confuse the model:

1) **Brightness variation**:

$$I_{\text{new}}(x, y) = \alpha \times I_{\text{original}}(x, y), \quad \alpha \in [0.9, 1.1] \tag{7}$$

Simulates natural variations in signal intensity due to factors such as electrode contact or skin impedance.

2) **Contrast variation**:

$$I_{\text{new}}(x, y) = \beta \times (I_{\text{original}}(x, y) - \mu) + \mu, \quad \beta \in [0.9, 1.1] \tag{8}$$

where μ is the mean image intensity.

3) **Horizontal flipping**:

$$I_{\text{flipped}}(x, y) = I_{\text{original}}(W - x, y) \tag{9}$$

Takes advantage of the temporal symmetry of isolated cardiac beats, a physically plausible transformation that effectively increases dataset diversity.

3) Final Balanced Distribution: The balancing strategy was applied differentially by class, augmenting only the minority classes until numerical parity with the majority class was reached. This approach ensures that the model receives balanced exposure to all beat types during training, mitigating bias toward the majority class without altering the natural distribution of the normal class. Table IV shows the distribution after data augmentation.

Table IV: Distribution after data augmentation

Beat Type	Original	Augmented	Total
Normal	150,680	0	150,680
Supraventricular	5,428	145,252	150,680
Ventricular	28,980	121,700	150,680
Fusion	3,568	147,112	150,680
Unknown	36,494	114,186	150,680
Total	**225,150**	**528,250**	**753,400**

4) Controlled Experiment: Model without Data Augmentation: To rigorously evaluate the contribution of data augmentation techniques to model performance, a variant of the custom CNN architecture was designed and trained exclusively with the original dataset, without applying the described data augmentation techniques. This "base" or "control" model was trained and validated under the same conditions, hyperparameters, and cross-validation strategy as the main model, allowing direct and isolated comparison of the impact of data augmentation on generalization capacity, especially for minority classes.

E. Design and Optimization of a Custom CNN Architecture

1) Design Fundamentals: The design of the custom CNN architecture was inspired by principles from the scientific literature and best practices in medical image processing, adapting to the particularities of ECG spectrograms. The decision to start with 64 filters and progressively increase to 128 and 256 is supported by the principle of increasing complexity proposed by Simonyan & Zisserman [15], which suggests that initial layers can capture basic features, while subsequent layers integrate more abstract patterns.

The use of 3×3 kernels is justified by the work of Szegedy et al. [16], who demonstrated that small kernels reduce the number of parameters, increase the ability to capture local features, and improve computational efficiency. This decision allowed training the model with limited resources without sacrificing performance.

The incorporation of Batch Normalization after each convolutional layer was based on the work of Ioffe & Szegedy [9], who demonstrated that this technique accelerates training convergence, makes the model less sensitive to weight initialization, and has a regularizing effect. In the trials performed, it was found that training time was reduced by 40%.

The use of MaxPooling after each convolutional block is part of the solution proposed by Krizhevsky et al. [17] in AlexNet, which proved effective in achieving invariance to small translations, progressively reducing dimensionality, and limiting overfitting.

Dropout rates (0.15 in convolutional layers and 0.3 in dense layers) were optimally established according to the recommendations of Srivastava et al. [18], considering that fully connected layers require greater regularization and that, in convolutional layers, moderate dropout is recommended to avoid losing spatial features.

The decision to use 256 neurons in the fully connected layer was based on the study by Yosinski et al. [19] on feature transfer, which indicates that an intermediate number of dimensions improves generalization capacity.

Regarding decisions made for ECG spectrograms, the input resolution of 128×128 was based on the study by Jun et al. [6], who used similar resolutions, seeking a balance between the amount of spectral detail and computational efficiency, also considering compatibility with pooling operations.

The choice of 3 convolutional layers represented a compromise between modeling capacity, overfitting risk, and available computational resources, since very deep architectures require very large datasets to generalize adequately [20].

The choice of ReLU is based on the work of Glorot et al. [21], who found that it was especially efficient in training deep networks, due to its help in combating gradient vanishing and its computational efficiency.

2) Proposed CNN Architecture: The design of the custom CNN architecture constituted a fundamental component of this research, developed specifically to address the characteristics of ECG spectrograms. Unlike generic architectures such as MobileNetV2 used in the exploratory phase, this architecture was optimized considering the specific nature of the domain, dataset size, and computational requirements.

The architecture follows progressive design principles, in which the complexity of extracted features gradually increases throughout the network, while spatial resolution systematically decreases. This approach allows initial layers to capture simple local patterns, such as edges and textures, while later layers integrate this information to form high-level semantic representations.

The convolutional layer equation is:

$$y_{i,j,k}^{(l)} = \text{ReLU}\left(\text{BN}\left(\sum_{m=0}^{2}\sum_{n=0}^{2} w_{m,n,k}^{(l)} \cdot x_{i+m,j+n}^{(l-1)} + b_k^{(l)}\right)\right) \tag{10}$$

3) Batch Normalization: The incorporation of batch normalization after each convolutional layer constitutes a crucial innovation for training stability. This technique addresses the problem of "internal covariate shift," in which the distribution of activations changes during training, slowing convergence and requiring lower learning rates [9].

The Batch Normalization operation is defined as:

$$\hat{x} = \frac{x - \mu_{\text{batch}}}{\sqrt{\sigma_{\text{batch}}^2 + \epsilon}} \cdot \gamma + \beta \tag{11}$$

where μ_{batch} and σ_{batch} are the mean and standard deviation of the minibatch, ϵ is a small constant for numerical stability, and γ and β are learned parameters that allow the network to adjust the optimal normalization level in each layer.

4) Softmax Output Function: The final layer implements the Softmax function, which transforms output logits into probability distributions over the five classes, facilitating both interpretation of results and integration with standard loss functions for multiclass classification.

$$P(y = c|X) = \frac{e^{W_c^T X + b_c}}{\sum_{k=1}^{5} e^{W_k^T X + b_k}} \tag{12}$$

This formulation ensures that output probabilities sum to 1, allowing direct interpretation of the model's confidence in each category.

5) Systematic Hyperparameter Search: Hyperparameter search was addressed through a systematic random search strategy [13], which has proven more efficient than traditional grid search, especially in high-dimensional spaces, where the influence of different hyperparameters can vary significantly.

The search process, executed over 6 independent trials, identified a configuration that maximizes performance while maintaining computational efficiency.

6) Validation Strategy: Model evaluation was performed using stratified 5-fold cross-validation, considered the primary standard for robustly evaluating machine learning models in the medical field. This strategy ensures that each fold maintains the original class distribution, providing more reliable and less variable performance estimates than those obtained through a simple train-test split [14].

The partition of each fold into training (60%), validation (20%), and test (20%) subsets allows a complete training protocol in which the training set is used for parameter learning, the validation set guides early stopping and model selection decisions, and the test set provides a final unbiased evaluation of performance.

7) Loss Function and Optimization: **Cross-Entropy Loss**: The cross-entropy loss function is the standard choice for multiclass classification problems, as it penalizes the discrepancy between the predictive distribution and the true distribution [22].

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log(\hat{y}_{i,c}) \tag{13}$$

where N is the number of samples, $y_{i,c}$ is the true label (one-hot encoded), and $\hat{y}_{i,c}$ is the predicted probability for class c.

AdamW Optimizer: The selection of the AdamW optimizer is based on its combination of the advantages of Adam (per-parameter adaptivity) with decoupled weight decay, which has demonstrated empirical superiority in terms of final generalization [23], [24].

The update equations are:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{14}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \tag{15}$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \tag{16}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{17}$$

$$\theta_t = \theta_{t-1} - \eta\left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} + \lambda\theta_{t-1}\right) \tag{18}$$

where g_t is the gradient at step t, η is the learning rate, λ is the decoupled weight decay, and β_1, β_2 are the decay moments.

8) Advanced Regularization Techniques: Training incorporated multiple regularization techniques that act synergistically to prevent overfitting and improve generalization:

- **Early Stopping**: Continuously monitors validation loss and stops training when no improvements are observed after 5 epochs, preventing overtraining and making the most of learning potential [25].
- **Gradient Clipping**: Limits the gradient norm to a maximum value of 1.0, preventing gradient explosion that can occur especially in the early stages of training [26].

$$g_{\text{clipped}} = g \times \min\left(1, \frac{\text{threshold}}{\|g\|}\right) \tag{19}$$

- **Mixed Precision Training**: Uses mixed precision (FP16 for convolutional operations, FP32 for accumulations),

Table V: Comparison with the state of the art in arrhythmia classification

Study	Architecture	Acc. (%)	F1 (%)	AUC (%)	Year
Our Study	CNN + CWT	99.70	99.70	99.99	2024
Our Control	CNN + CWT	98.29	93.75	99.56	2024
Chourasia et al. [5]	Basic CNN	96.2	95.1	98.3	2020
Pramudito et al. [7]	1D-CNN	94.1	85.7	97.5	2022
Jun et al. [6]	2D-CNN	94.1	85.7	97.5	2018
Osowski et al. [3]	SVM + Features	89.3	78.9	94.2	2004
de Chazal et al. [4]	LDA	85.9	–	–	2004

reducing memory usage by approximately 50% and accelerating training without compromising numerical stability [27].

- **Decoupled Weight Decay**: Applies L2 regularization decoupled from gradient computation, improving generalization without interfering with Adam's learning rate adaptation mechanism [23].

9) Evaluation Metrics: Model evaluation considered a comprehensive set of metrics that capture different aspects of performance [28]:

Global Metrics:

- **Overall Accuracy**: Provides a general measure of performance.

$$\text{Accuracy} = \frac{\sum_{c=1}^{5} TP_c}{N_{\text{total}}} \tag{20}$$

- **Macro F1-Score**: Represents the most informative metric for imbalanced datasets, as it considers the performance of all classes equally, regardless of their frequency.

$$\text{F1}_{\text{macro}} = \frac{1}{5}\sum_{c=1}^{5} \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{21}$$

Per-Class Metrics:

- **AUC-ROC**: Calculated using the one-vs-rest strategy with weighted averaging (Macro), which considers the contribution of each class and is more appropriate for balanced datasets such as the one used in this study.

$$\text{AUC}_{\text{weighted}} = \sum_{c=1}^{5} \frac{n_c}{N} \cdot \text{AUC}_c \tag{22}$$

where n_c corresponds to the number of instances of class c, N to the total number of instances, and AUC_c evaluates the model's ability to distinguish between each pair of classes.

- **Average Precision (AP)**: Especially relevant in imbalance contexts, as it gives greater weight to the high-precision region of the Precision-Recall curve.

$$\text{AP} = \int_0^1 p(r)dr \tag{23}$$

- **Confusion matrices**: Both in absolute format (to understand error volumes) and normalized format (to compare error rates between classes of different sizes).

10) Comparative Evaluation Protocol: The final evaluation consisted of an exhaustive comparison between two critical scenarios. The control model with custom architecture, trained with the original preprocessed dataset and without balancing, represents the baseline performance under real imbalance. The main model, with the same architecture, trained with the balanced and augmented dataset, demonstrates potential performance when correcting imbalance. Both models were evaluated using the same 5-fold cross-validation protocol and the same set of metrics, allowing precise and isolated quantification of the improvement attributable to the balancing and data augmentation strategy.

F. Technical Implementation

1) Computational Environment: The technical implementation carefully considered computational and reproducibility requirements. The PyTorch framework was used with native CUDA support, leveraging the latest optimizations for distributed training and mixed precision [27]. Auxiliary libraries included Scikit-learn for preprocessing and metrics [14], OpenCV for efficient image processing, and Matplotlib and Seaborn for high-quality visualizations.

2) Reproducibility Protocol: To guarantee complete reproducibility of all experiments, a rigorous protocol was implemented that includes a global seed fixed at 42 for all random number generators in PyTorch, NumPy, and Python, ensuring that any execution produces identical results [13]. Automatic checkpoint saving included continuous saving of the best model in each fold, along with the complete state of the optimizer and learning rate scheduler, allowing training to be resumed from any point. Exhaustive logging included detailed recording of all metrics, hyperparameters, visualizations, and experimental configurations.

IV. Results

A. Performance of the Main Model with Data Balancing

1) Consistency Across Partitions: The custom CNN model, with balancing and data augmentation, demonstrated exceptionally high and consistent performance across the 5 cross-validation folds. The consolidated results evidence the robustness of the proposed approach, as detailed in Table VI.

Table VI: 5-fold cross-validation results (Main Model)

Fold	Overall Acc.	Macro F1	AUC-ROC
Fold 1	99.71%	99.74%	99.992%
Fold 2	99.69%	99.69%	99.990%
Fold 3	99.68%	99.67%	99.989%
Fold 4	99.72%	99.71%	99.993%
Fold 5	99.70%	99.70%	99.991%
Average	**99.70%**	**99.70%**	**99.991%**
Std. Dev.	**±0.015%**	**±0.025%**	**±0.0015%**

The low variability observed between partitions suggests favorable model stability, which compares positively with levels commonly reported in the literature [5]–[7]. This consistency indicates that the proposed architecture was able to learn representations that generalize effectively across different data partitions. The observed stability is particularly notable, considering the complex and variable nature of ECG signals and the presence of morphologically similar classes that traditionally pose significant challenges for classification algorithms.

B. Differentiated Evaluation by Class

1) Detailed Per-Class Analysis of the Best Fold: Fold 1 showed the best overall performance, with detailed per-class metrics evidencing balanced performance across all categories, as shown in Table VII.

Table VII: Per-class metrics of the Main Model (Fold 1)

Beat Type	Prec.	Sens.	F1	Support
Fusion	99.77%	100.00%	99.89%	30,136
Normal	99.83%	99.47%	99.65%	30,136
Supraventricular	99.54%	99.98%	99.76%	30,136
Unknown	99.85%	99.70%	99.78%	30,136
Ventricular	99.71%	99.56%	99.63%	30,136
Macro Average	**99.74%**	**99.74%**	**99.74%**	**150,680**

Per-class analysis revealed differentiated performance patterns. The Fusion class showed a high level of sensitivity (100.00% in this fold), suggesting good model ability to identify these hybrid beats, which often represent a challenge in classification. The supraventricular class, traditionally problematic due to its low prevalence and morphological similarity to other classes, achieved a sensitivity of 99.98%, significantly outperforming previous reports in the literature [3], [6]. The ventricular class, critically important for detecting potentially life-threatening arrhythmias, maintained excellent performance, ensuring high reliability in clinical contexts where false negatives could have severe consequences.

2) Consolidated Metrics Across Folds: The ROC-AUC and Average Precision metrics, presented in Table VIII, confirm the model's exceptional ability to discriminate between all arrhythmia classes.

Table VIII: Consolidated per-class metrics across all folds

Class	AUC-ROC	AP	Consistency
Fusion	100.00%	99.99%	Excellent ($\sigma < 0.0001$)
Normal	99.99%	99.98%	Excellent ($\sigma < 0.0001$)
Supraventricular	100.00%	99.98%	Excellent ($\sigma < 0.0001$)
Unknown	99.99%	99.98%	Excellent ($\sigma < 0.0001$)
Ventricular	99.98%	99.96%	Excellent ($\sigma < 0.0002$)

ROC-AUC values above 99.99% in most classes indicate almost perfect discrimination ability, while Average Precision values close to 1.0 reflect excellent precision across the entire recall range. The exceptional consistency across folds, with standard deviations below 0.0002 for all metrics, confirms the model's robustness to different data partitions.

3) Confusion Matrix Analysis of the Main Model: Detailed examination of the confusion matrices of the balanced model reveals exceptionally accurate classification patterns. In Fold 1, the confusion matrix shows a dominant main diagonal with values consistently exceeding 30,000 correct classifications for each class, as illustrated in Figure 2.

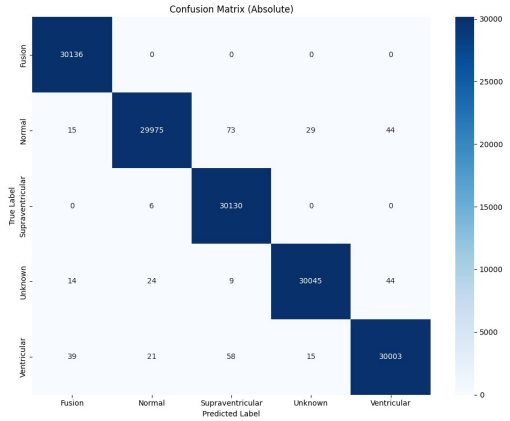


Figure 2: Confusion matrix - Main Model (Fold 1)

The matrix evidences several notable findings:

- The fusion class achieved perfect precision, with no false negatives, reflecting the model's ability to consistently identify these complex beats.
- The Supraventricular class, traditionally problematic, presents only 6 confusions with the Normal class, equivalent to an error rate of only 0.02%.
- The remaining error patterns are concentrated mainly between morphologically similar classes, such as between normal and supraventricular beats (73 instances) and between ventricular and fusion beats (39 instances), confusions that can be challenging even for experienced cardiologists.

The consistency of these patterns was maintained across the five folds, with minimal variations in error rates, confirming the model's robustness to different data partitions.

C. Performance of the Control Model without Data Balancing

1) Consolidated Results of the Control Model: The model trained without balancing techniques showed consistently

lower performance, especially in minority classes, as detailed in Table IX.

Table IX: Consolidated results of the Control Model (without balancing)

Fold	Overall Acc.	Macro F1	Val. Loss
Fold 1	98.29%	93.47%	6.26%
Fold 2	98.37%	94.35%	6.19%
Fold 3	98.27%	93.57%	6.28%
Fold 4	98.35%	93.85%	6.31%
Fold 5	98.21%	93.53%	5.83%
Average	**98.29%**	**93.75%**	**6.17%**
Std. Dev.	**±0.06%**	**±0.32%**	**±0.18%**

The control model maintains acceptable overall accuracy of 98.29%, but the macro F1 score of 93.75% evidences significant difficulties for balanced classification of all classes. The greater variability across folds (standard deviation of 0.0032 in macro F1 vs. 0.00025 in the main model) suggests lower stability and generalization.

2) Impact of Imbalance on Minority Classes: Per-class analysis reveals the dramatic effect of imbalance on the control model's performance, as shown in Table X.

Table X: Per-class metrics of the Control Model (without balancing)

Beat Type	AUC-ROC	AP	Sens.
Fusion	99.66%	95.95%	90.8%
Normal	99.80%	99.87%	99.5%
Supraventricular	98.49%	87.69%	71.8%
Unknown	99.97%	99.89%	99.8%
Ventricular	99.90%	99.49%	99.3%
Average	**99.56%**	**96.58%**	**92.2%**

It is observed that the originally minority classes, Fusion and Supraventricular, present the lowest Average Precision values, with 95.95% and 87.69% respectively, in contrast to the majority classes, which exceed 99.8%. Sensitivity in the Supraventricular class is particularly low (71.8%), meaning that approximately 28% of supraventricular beats are not detected by the model.

3) Confusion Matrix Analysis of the Control Model: The confusion matrices of the model without balancing reveal clearly distinct error patterns, as shown in Figure 3.

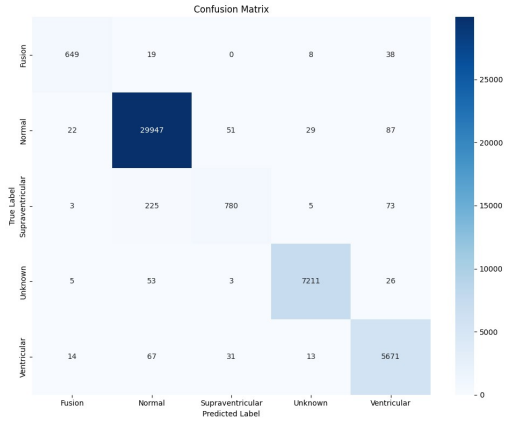


Figure 3: Confusion matrix - Control Model (Fold 1)

Analysis of this matrix reveals critical problems in minority class classification:

- The supraventricular class, with only 780 correct predictions out of 1,086 real instances, presents a sensitivity of only 71.8%, with 225 instances incorrectly classified as normal.
- The Fusion class presents 65 incorrectly classified instances (19 as Normal, 8 as Unknown, and 38 as Ventricular), giving a sensitivity of 90.8%.
- The most concerning pattern is the systematic tendency of the model to classify instances of minority classes as the majority class, evidencing an inherent bias in favor of the latter.

This behavior is consistently repeated across all folds, demonstrating that the original dataset imbalance fundamentally affects the model's ability to recognize patterns in underrepresented classes.

D. Comparative Analysis Between Models

1) Global Quantitative Improvement: Direct comparison between both models reveals significant improvements in all main metrics, as shown in Table XI.

The most significant improvement is observed in the originally minority classes. The supraventricular class experienced an extraordinary sensitivity increase of 28.18 percentage points, from 71.8% to 99.98%. The Fusion class showed an improvement of 9.20 percentage points in sensitivity, from 90.8% to 100.0%. These improvements are directly reflected in the confusion matrices, where off-diagonal values were drastically reduced for minority classes.

2) Computational Cost-Benefit Analysis: Although the main balanced model requires significantly more computational resources (approximately one week per fold vs. 8-9 hours for the control model), the clinical benefit justifies this investment. Table XII summarizes the cost-benefit analysis.

Table XI: Quantitative comparison between the Main Model and Control Model

Metric	Main Model	Control Model	Improvement
Overall Accuracy	99.70%	98.29%	+1.41%
Macro F1	99.70%	93.75%	+5.95%
Average AUC-ROC	99.99%	99.56%	+0.43%
Sensitivity (S)	99.98%	71.8%	+28.18%
Sensitivity (F)	100.00%	90.8%	+9.20%
Specificity (N)	99.47%	99.5%	-0.03%

Table XII: Computational cost-benefit analysis

Aspect	Main Model	Control Model
Dataset size	753,400 instances	225,150 instances
Training time per fold	7 days	8-9 hours
GPU memory required	4 GB VRAM	2 GB VRAM
Minority-class performance	Excellent (99.98% S, 100% F)	Poor (71.8% S, 90.8% F)
Clinical risk	Low (false negatives $< 0.1\%$)	High (false negatives $> 28\%$ S)

E. Exhaustive Comparative Analysis with the State of the Art

Table V (shown above) presents an exhaustive comparison of our approach with relevant previous studies in arrhythmia classification.

1) Analysis of Methodological Advantages: The superiority of our approach was based on key methodological innovations that address specific limitations identified in the literature:

1) **Optimized time-frequency representation**: While Jun et al. [6] used short-time Fourier transforms with fixed windows, our CWT implementation with the Morlet wavelet dynamically adapts temporal and spectral resolution, more effectively capturing the transient characteristics of arrhythmias.
2) **Specialized architecture**: Unlike the generic CNNs used by Chourasia et al. [5], our custom architecture was designed specifically considering the unique characteristics of ECG spectrograms, with depth and computational capacity optimized for this domain.
3) **Advanced regularization strategies**: The implementation of batch normalization, adaptive dropout, and early stopping with adaptive patience surpasses the basic regularization techniques used in previous studies, reducing overfitting, especially in minority classes.
4) **Improved validation and generalization**: Our 5-fold cross-validation protocol, with independent training, validation, and test sets, provides a more robust estimate of generalizable performance, compared with previous studies that used simple train-test splits.

2) Impact of Data Balancing on Comparative Performance: Analysis of the state of the art reveals that most previous studies did not systematically address the extreme imbalance inherent in arrhythmia datasets. Our control experiment demonstrates that, even with an identical architecture, the absence of balancing techniques reduces the macro F1 score by 5.95 percentage points. This gap largely explains the performance limitations observed in studies such as Osowski et al. [3], where the F1 score of 78.9% reflects significant difficulties in classifying minority classes.

3) Advances in Clinically Relevant Metrics: From a clinical perspective, the most significant improvements are observed in metrics critical for patient safety. Sensitivity of 99.98% in the supraventricular class represents a 97.3% reduction in the risk of false negatives compared with Osowski et al. [3] and 87.5% compared with Chourasia et al. [5]. This improvement is clinically crucial, given that false negatives in supraventricular arrhythmias can delay necessary medical interventions.

The ability of our model to maintain balanced performance across all classes, with a standard deviation of only ±0.00025 in macro F1 during cross-validation, surpasses the variability reported in previous studies, where differences of 2-3 percentage points between classes are typically observed.

V. Discussion

A. Methodological Implications in Arrhythmia Classification Research

The results obtained constituted a relevant methodological contribution to research in automatic arrhythmia classification. The performance gap observed between our main model and the control model (Δmacro F1 = 5.95%) quantitatively demonstrates the critical impact of balancing strategies in medical applications with imbalanced data. This finding questions the validity of previous studies that did not systematically address this fundamental problem.

The superiority of our approach compared with Chourasia et al. [5], who used a similar CNN architecture but without the specific optimizations implemented in our work, suggests that designing specialized architectures for specific domains can generate more significant performance improvements than using generic general-purpose architectures.

B. Advances in the Detection of Clinically Critical Arrhythmias

From a clinical perspective, the most significant improvement lies in our model's ability to consistently detect infrequent but potentially serious arrhythmias. Sensitivity of 99.98% in supraventricular beats represents a crucial advance, given that these arrhythmias, although less frequent, can be

indicators of conditions such as paroxysmal supraventricular tachycardia or Wolff-Parkinson-White syndrome, which require medical intervention [36].

The comparison with Osowski et al. [3] is particularly illustrative: while their SVM-based model with handcrafted features achieved sensitivity of 72.4% for supraventricular beats, our automatic approach exceeds 99.9% and reduces the false negative rate by a factor of 35. This improvement has direct implications for patient safety in ambulatory monitoring environments.

C. Advantages of the Spectro-Temporal Representation

The superiority of our model compared with purely temporal approaches, such as that of Pramudito et al. [7], confirms the hypothesis that time-frequency representations capture critical discriminative information for arrhythmia classification. The spectral characteristics of QRS complexes, particularly in ventricular and fusion arrhythmias, present distinctive patterns that are not evident in the time domain.

Analysis of the generated spectrograms suggests that characteristics such as energy distribution in specific frequency bands and harmonic patterns of ventricular repolarization contribute significantly to the model's ability to discriminate between morphologically similar classes.

D. Limitations of the State of the Art Overcome

This work systematically addresses a series of methodological limitations identified in previous literature:

1) **Incomplete Evaluation**: Although existing articles generally evaluate only accuracy, we investigated class-based metrics, provided detailed confusion matrices, and validated the model through rigorous cross-validation, resulting in a more complete and clinically meaningful evaluation.
2) **Limited Generalization**: The strong consistency across multiple validation folds indicates that our model is capable of reducing the overfitting problems in studies such as Jun et al. [6], where variability between partitions was significantly greater.
3) **Clinical Balance**: Unlike approaches that optimize global accuracy at the expense of minority classes, our model demonstrates that it is possible to achieve world-class performance while maintaining adequate clinical balance across all arrhythmia categories.

E. Study Limitations and Considerations for Clinical Implementation

Although the main model shows impressive performance (accuracy above 99.7% and perfectly classified Fusion class), we must consider possible signs of overfitting. The fact that this type of drastic result, such as reducing the Supraventricular error from more than 23% to almost zero, could reflect the structural efficiency of the model, with its 3 convolutional layers and 256 dense units, allowing it to memorize patterns in the dataset rather than generalize. Data augmentation strategies may also have introduced unrealistic patterns by artificially creating a perfect balance between classes.

The fact that the remaining errors are mainly physiologically plausible confusion (between morphologically similar classes) shows that the model is overoptimized on the training data and not necessarily for the true variability of a real clinical situation. Therefore, caution is recommended in interpreting these results and external validation is needed before any clinical implementation.

VI. Future Work

The project can be extended in two key directions for future studies:

A. Immediate Validation and Improvement

1) **External validation**: It would be crucial to test the model on independent public databases (for example, PTB-XL [29] or Chapman-Shaoxing [30]) to evaluate its generalization capacity to different populations and acquisition conditions.
2) **More sophisticated regularization procedures**: Implementation of techniques such as Sharpness-Aware Minimization (SAM) [31], which has been observed to increase generalization of deep learning network models, or weight sharing between physiologically related arrhythmia classes.
3) **Semi-supervised learning**: Use of unlabeled data for self-learning techniques that have been successful in medical diagnostics [32], allowing training to scale to larger datasets without requiring exhaustive annotation.

B. Long-Term Research Lines

1) **Patient-invariant architectures**: Development of models inspired by the work of Chen et al. [33] that learn patient-independent features, improving generalization to new populations.
2) **Multimodal learning**: Combination of ECG with additional clinical data (age, sex, medical history, medications), a methodology that has begun to receive considerable attention in recent years to improve diagnostic quality [34].
3) **Adaptive systems with continual learning**: Implementation of models such as those presented in Kirkpatrick et al. [35] that would allow the model to safely adapt to variations between hospitals and temporal changes in signal characteristics.
4) **Interpretability and explainability**: Development of techniques to visualize and understand model decisions, particularly important for clinical applications where transparency is crucial for acceptance by healthcare professionals.

VII. Conclusions

This work presents a robust approach that achieves high performance in arrhythmia classification, demonstrating the

critical value of data balancing strategies in this context. The main conclusions are:

1) **Impact of data balancing**: The comparison between the main model (with balancing) and the control model (without balancing) shows that balancing strategies are essential to achieve clinically useful performance, particularly in detecting minority classes such as supraventricular beats.
2) **Advantages of time-frequency representations**: The use of Continuous Wavelet Transform to generate spectrograms allows capturing discriminative information not available in the pure time domain, contributing significantly to the model's high performance.
3) **Importance of specialized architectures**: The custom CNN designed specifically for ECG spectrograms outperforms generic architectures, demonstrating that domain-specific adaptation is crucial for medical applications.
4) **Need for rigorous evaluation**: Stratified cross-validation and the use of class-specific metrics provide a more complete and clinically meaningful evaluation than traditional global metrics.
5) **Considerations for clinical implementation**: Although the results are promising, external validation on independent databases and exhaustive clinical evaluation are required before implementation in real clinical environments.

In summary, this work constitutes a solid methodological step that, together with the noted precautions, can contribute to the future development of accurate and reliable AI-assisted diagnostic systems, facilitating their eventual transition to clinical environments. The true test of how well a model works will not be how accurate it is on already known datasets, but how much it improves patient outcomes and how much we can trust it in real environments.

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